

GAS LEAKAGE ALARM

■ SANI THEO

LPG gas is supplied in pressurised steel cylinders. As this gas is heavier than air, when it leaks from a cylinder it flows along floor and tends to settle in low spots such

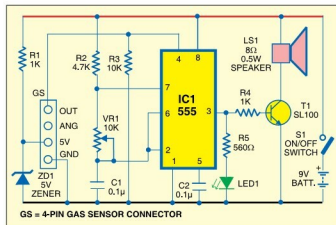


Fig. 1: Circuit for gas leakage alarm

as a basement. This can cause fire or suffocation if not dealt with.

Here is a circuit that detects the leakage of LPG gas and alerts the user through audio-visual indications.

Fig. 1 shows the circuit of the gas leakage alarm. The circuit operates off a 9V PP3 battery. Zener diode ZD1 is used to convert 9V into 5V DC to drive the gas sensor module.

The SEN-1327 gas sensor module from RhydoLABZ is used in this circuit. Its output goes high when the gas level reaches or exceeds certain point. A preset in the module is used to set the threshold. Interfacing with the sensor module is done through a 4-pin SIP header.

Pin details of the gas sensor module are

shown in Fig. 2. An MQ-6 gas sensor is used in the gas sensor module. As per its datasheet, it has high sensitivity to propane, butane, isobutene, LPG and natural gas. The sensor can

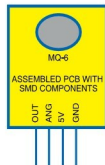


Fig. 2: Pin details of gas sensor module

also be used to detect combustible gases, especially methane. This circuit has been tested with LPG gas and was found to work satisfactorily.

Whenever there is LPG concentration of 1000 ppm (parts per million) in the area, the OUT pin of the sensor module goes high. This signal drives timer IC 555, which is wired as an astable multivibrator. The multivibrator basically works as a tone generator.

Output pin 3 of IC 555 is connected to LED1 and speaker-driver transistor SL100 through current-limiting resistors R5 and R4, respectively. LED1 glows and the alarm sounds to alert the user of gas leakage. The pitch of the tone can be changed by varying preset VR1. Use a suitable heat-sink for transistor SL100. ●

ELECTRONIC ELECTROSCOPE

■ RAJU BADDI

An electroscope is the instrument used to detect charged bodies. Here is an electronic version of the scope that is sensitive and, unlike conventional scopes, indicates the polarity of charge as well. This circuit consumes very low quiescent power and reliably indicates charge induction and detection. The polarity of charge is indicated through LEDs (green LED indicates positive and red LED negative).

As shown in Fig. 1, the circuit is built around popular dual timer 556 (IC1), quad 2-input NAND gate CD4011 (IC2), transistors BC547 and BC557, and a few discrete components. The CD4011 has four NAND gates with very high input impedance. This high impedance is used in detecting the potential difference produced by an external charge distribution.

When the positively charged object is brought near the positive-detecting plate, it attracts electrons from the detector plate and hence capacitor C6

becomes positively charged.

Diodes D3 and D4 (1N4148) help determine the direction of current flow, removing certain shortcomings of the circuit. (These prevent the capacitor of the opposite arm from charging when the object is withdrawn quickly.) Even though diode D4 in series with capacitor C6 is reverse biased, the current through it is adequate to charge capacitor C6. Thus capacitor C6 gets charged with respect to the ground